

# Multiplicity Operator Calculus Resonance Classes & Proof-by-Resonance

**Goal:** Formalize MOC (prime-indexed noncommutative operator calculus on signals + hypergraph relations) with:

1. **Resonance classes** as near-optimal moduli (clustered maximizers), and
2. **Proof-by-resonance** as a reproducible certificate protocol, including an explicit **vector resonance**

$$R = (R_1, R_2, R_3) \in [0, 1]^3$$

with (i) a **topology-sensitive** term for hypergraph rewrites and (ii) **null-model calibration** giving the acceptance threshold  $\tau_R$  statistical “teeth.”

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## 1. Core objects

### 1.1 Time lattice and signals

Let  $n \in \mathbb{N}$  and  $T_n := \mathbb{Z}/n\mathbb{Z}$ .

Signal space:

$$X_n = \{x: T_n \rightarrow \mathbb{R}^k\}.$$

Write  $x(t) \in \mathbb{R}^k$  and use the inner product  $\langle u, v \rangle$  on  $\mathbb{R}^k$ .

### 1.2 Hypergraph relations

A (possibly weighted) hypergraph is

$$H = (V, E, \iota, w),$$

where  $\iota(e) \subseteq V$  is the incidence of hyperedge  $e$ , and  $w: E \rightarrow \mathbb{R}_{\geq 0}$  are edge weights.

We allow **state attachments** on vertices and/or edges. For concreteness, let vertex states be derived from the signal:

$$y_v(t) := \Phi_v(x(t)) \in \mathbb{R}^d \quad (\forall v \in V),$$

for some fixed feature map  $\Phi_v$ .

### 1.3 Multiplicity space

The MOC “state of the world” is

$$\mathcal{M} = (X_n, H).$$

A dataset  $D$  provides observed signals and (optionally) observed relational events. We treat  $D$  as the evaluation anchor for resonance.

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## 2. Prime-indexed operator calculus (sketch)

Let  $\Sigma$  be an operator alphabet containing prime-indexed families:

- Subdivision  $S_p$
- Accent / gating  $A_{p^r}$
- Rotation  $\text{Rot}_{p^r}$
- Within-cell permutation  $W_p$
- Relation/topology operators  $\hat{P}$  acting on  $H$  (split/merge/fold/relabel, etc.)

A **word** (program) is an ordered product

$$W = \hat{O}_m \cdots \hat{O}_2 \hat{O}_1 \in \Sigma^*.$$

Noncommutativity is essential: permuting operators generally changes outcomes.

Each word acts on  $\mathcal{M}$  with (optional) continuous parameters  $\theta \in \Theta$ :

$$(W, \theta): \mathcal{M} \mapsto \mathcal{M}' = (X_n', H').$$

**Admissibility / invariants.** Let  $\text{Inv}(W, \theta) = \text{true/false}$  be the conjunction of all declared invariants (fairness budgets, topology budgets, energy budgets, etc.). Only admissible  $(W, \theta)$  are considered.

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## 3. Gauge-fix (canonicalization)

Let  $\text{gf}$  be a **deterministic canonicalization** procedure that removes internal symmetries (e.g., relabeling, phase pinning) to obtain a canonical representative.

We will evaluate resonance on the gauge-fixed representative:

$$\bar{R}(W, \theta; D) := R(\text{gf}(W), \theta; D).$$

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We define three components, each in  $[0, 1]$ . The overall scalar score can be any monotone aggregator (weighted sum, geometric mean). We recommend geometric mean to penalize “one-metric hacks.”

**Idea:** If MOC finds genuine  $p^r$ -tier structure, the transformed signal should be coherent when projected onto the  $p^r$ -grid.

For each tier  $q = p^r \mid n$ , define a tier projector  $\Pi_q$  that aggregates time points by residue class modulo  $q$ :

$$(\Pi_q x)(a) = \frac{1}{n/q} \sum_{t \equiv a \pmod{q}} x(t), \quad a \in T_q.$$

Let  $x_W$  be the signal after applying  $(W, \theta)$ . Define a normalized tier energy

$$E_q(x_W) = \frac{\sum_{a \in T_q} |(\Pi_q x_W)(a)|^2}{\sum_{t \in T_n} |x_W(t)|^2 + \epsilon} \in [0, 1].$$

Let  $\mathcal{Q}$  be the set of tiers under consideration (e.g., all  $p^r \mid n$  up to a max exponent). Define

$$R_1(W, \theta; D) = \sum_{q \in \mathcal{Q}} \alpha_q E_q(x_W), \quad \alpha_q \geq 0, \sum_q \alpha_q = 1.$$

Interpretation: high  $R_1$  means the word induces coherent tier structure across prime grids.

**Idea:** Vertex states derived from the transformed signal should agree with the hypergraph relations (e.g., vertices linked by hyperedges should exhibit compatible features).

Let  $y_v$  be derived features (possibly time-averaged or tier-averaged), and let  $\psi$  be a compatibility function over hyperedges:

$$\psi_e(\{y_v\}_{v \in \iota(e)}) \in [0, 1].$$

Examples:

- agreement:  $\psi_e = \exp(-\text{Var}_{v \in \iota(e)} \|y_v\|)$
- constraint satisfaction:  $\psi_e = \mathbf{1}[\text{constraint satisfied}]$
- learned edge score:  $\psi_e = \sigma(\text{MLP}(\text{pool}(y_v)))$

Define edge-aggregated fit

$$R_2(W, \theta; D) = \frac{\sum_{e \in E} w(e) \psi_e(\{y_v\}_{v \in \iota(e)})}{\sum_e w(e) + \epsilon}$$

Interpretation: high  $R_2$  means the signal-induced states are compatible with relational structure.

We need  $R_3$  to detect when relation-operators  $\hat{P}$  “cheat” by rewriting the hypergraph into a different causal story that nonetheless scores well on  $R_1, R_2$ .

We introduce a **topology distance** between the original hypergraph  $H$  and the rewritten  $H_W$ :

$$\Delta_{\mathrm{top}}(H, H_W) \geq 0.$$

Then define

$$R_3(W, \theta; D) = \exp(-\beta \Delta_{\mathrm{top}}(H, H_W)) \cdot \mathbf{1}[\text{rewrite budgets satisfied}].$$

Use a convex combination:

$$\Delta_{\mathrm{top}} = \gamma_1 \Delta_{\mathrm{spec}} + \gamma_2 \Delta_{\mathrm{inc}} + \gamma_3 \Delta_{\mathrm{edit}}, \quad \gamma_i \geq 0, \quad \sum \gamma_i = 1.$$

**(a) Spectral distance  $\Delta_{\mathrm{spec}}$** . Choose a hypergraph operator  $\mathcal{L}(H)$  (e.g., a normalized incidence-based Laplacian or application-specific diffusion operator). Let  $\lambda_1 \leq \dots \leq \lambda_K$  be the first  $K$  eigenvalues. Define

$$\Delta_{\mathrm{spec}}(H, H_W) = \frac{1}{K} \sum_{i=1}^K \frac{|\lambda_i(H) - \lambda_i(H_W)|}{|\lambda_i(H)| + \epsilon}.$$

This penalizes global structural drift.

**(b) Incidence-statistics distance  $\Delta_{\mathrm{inc}}$** . Let  $d(v)$  be vertex degree (counting hyperedge memberships), and  $s(e) = |\iota(e)|$  hyperedge size. Let  $P_H$  be the empirical distribution of degrees and sizes (or their histograms). Define

$$\Delta_{\mathrm{inc}}(H, H_W) = \mathrm{JS}(P_H, P_{H_W}) \in [0, \log 2],$$

where JS is Jensen–Shannon divergence (normalized to  $[0, 1]$  if desired).

**(c) Edit cost  $\Delta_{\mathrm{edit}}$** . Assign costs to rewrite primitives (split, merge, fold, relabel) and sum them:

$$\Delta_{\mathrm{edit}}(H, H_W) = \sum_{\text{ops}} \hat{P}(\text{used}) c(\text{op}).$$

This makes “more rewriting” explicitly expensive.

## 5. Scalar resonance and acceptance threshold

### 5.1 Scalar aggregator (recommendation)

Define scalar resonance

$$R_{\mathrm{sc}}(W, \theta; D) = \Big(R_1, R_2, R_3\Big)^{1/3}.$$

(Alternative:  $\sum_i \lambda_i R_i$ , but geometric mean discourages one-component overfitting.)

We define a **null distribution** for  $R_{\text{sc}}$  under structure-destroying transformations that preserve superficial marginals.

### 5.2.1 Null family

Construct null datasets  $D^{(b)}$  by combining one or more of:

1. **Time permutation within tiers:** shuffle time indices within each  $p^r$ -cell (preserves local statistics but destroys cross-cell alignment).
2. **Phase randomization:** randomize Fourier phases while preserving magnitude spectrum (for stationary assumptions).
3. **Hypergraph configuration null:** rewire incidence while preserving degree sequence and hyperedge size distribution (destroys higher-order relational motifs).
4. **Coupling break:** keep signal and hypergraph but randomly permute the mapping  $v \mapsto \Phi_v$  (destroys signal-relation alignment).

Let  $\mathcal{N}$  denote the null generator producing  $B$  samples  $\{D^{(b)}\}_{b=1}^B$ .

### 5.2.2 Search-matched calibration

Because MOC performs optimization over  $(W, \theta)$ , calibration must match the search procedure.

Let  $\mathcal{W}$  be the grammar/constraint set. Let  $\mathcal{A}$  be the fixed search algorithm (beam/MCTS) returning the best score on a dataset:

$$\hat{R}(D) := \max_{(W, \theta) \in \mathcal{A}(\mathcal{W}; D)} R_{\text{sc}}(W, \theta; D).$$

Compute null best-scores  $\hat{R}(D^{(b)})$ . Define

$$\tau_R := \text{Quantile}_{1-\alpha}(\{\hat{R}(D^{(b)})\}^B).$$

Then accepting a proof requires  $\hat{R}(D) \geq \tau_R$  (and the robustness checks below).

**Interpretation:**  $\tau_R$  is an optimization-aware significance threshold: it controls false discovery under the exact search power you used.

## 6. Resonance classes as near-optimal moduli

### 6.1 Near-optimal set

Let

$$R^*(D) = \sup_{(U, \phi) \in \mathcal{W}} \bar{R}_{\text{sc}}(U, \phi; D).$$

Use a  $\delta$ -optimal set:

$$\mathcal{R}_{\delta}(D) = \{(W, \theta) \in \mathcal{W} : \bar{R}(W, \theta; D) \geq R^*(D) - \delta\}.$$

## 6.2 Adjacency and class construction (clustered moduli)

Define an adjacency relation  $\rightsquigarrow$  on  $\mathcal{R}_{\delta}(D)$  by **one local edit**:

- swap adjacent noncommuting operators,
- apply one permitted rewrite primitive to  $H$  within budgets,
- small  $\theta$ -perturbation,
- or replace an operator by a declared neutral element.

Keep an edge only if

$$|\bar{R}_{\{\text{sc}\}}(W, \theta; D) - \bar{R}_{\{\text{sc}\}}(W', \theta'; D)| \leq \epsilon.$$

Define **resonance classes** as connected components of this graph. Denote the set of classes by

$$\mathcal{R}_{\delta, \epsilon}(D).$$

## 6.3 Class descriptors (what counts as “the explanation”)

For a class  $C \in \mathcal{R}_{\delta, \epsilon}(D)$ , define:

- **Tier signature:**  $\mathbf{e}(C) = \mathbb{E}_{(W, \theta) \sim C}[\{E_q(x_W)\}_{q \in \mathcal{Q}}]$
- **Order motifs:** empirical precedence probabilities  $P(\hat{O}_i \prec \hat{O}_j \mid C)$
- **Rewrite footprint:**  $\mathbb{E}[\Delta_{\text{top}}]$  and edit-type counts

These are the human-facing explanation artifacts.

# 7. Proof-by-resonance certificate protocol

A certificate for claim  $C$  is:

$$\Pi = (\mathcal{W}, \mathcal{A}, \text{gf}, \alpha, \tau_R, \delta, \epsilon, \Delta, \mathcal{T}, \text{logs})$$

plus a returned witness  $(W, \theta)$  and its class  $C_W \in \mathcal{R}_{\delta, \epsilon}(D)$ , satisfying:

1. **Significant score:**  $\hat{R}(D) \geq \tau_R$  under null-calibrated threshold.
2. **Ablation necessity:** for each declared essential operator  $\hat{O}$  in  $W$ ,  $|\bar{R}_{\{\text{sc}\}}(W \setminus \hat{O}; D) - \bar{R}_{\{\text{sc}\}}(W; D)| \leq \Delta$ .
3. **Held-out stability:** for transforms  $T \in \mathcal{T}$  (windows, mild noise, allowed relabelings),  $|\bar{R}_{\{\text{sc}\}}(W; T(D)) - \bar{R}_{\{\text{sc}\}}(W; D)| \leq \tau_R$  and class descriptors stay stable.
4. **Reproducibility:** deterministic evaluation + recorded seeds + exact grammar budgets and null generator settings.

## 8. Minimal implementation plan (first build)

1. Fix  $n$ , primes/tiers  $\mathcal{Q}$ , and a small grammar  $\mathcal{W}$  (bounded length, bounded rewrite budgets).
  2. Implement  $R_1$  via tier projectors  $\Pi_q$ .
  3. Implement  $R_2$  with a simple  $\psi_e$  (variance-based agreement or constraint satisfaction).
  4. Implement  $R_3$  with edit cost + incidence divergence; add spectral term later if needed.
  5. Implement search  $\mathcal{A}$  (beam) returning  $\hat{R}(D)$  and near-optimal set  $\text{Opt}_\delta(D)$ .
  6. Build resonance-class graph and compute connected components.
  7. Build null generator  $\mathcal{N}$  and compute  $\tau_R$  at level  $\alpha$ .
  8. Run negative controls (wrong  $n$ , forced commutativity, disabled rewrites) and verify collapse.
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## 9. Expected outcomes / diagnostics

- True multi-tier structure  $\Rightarrow$  multiple near-optimal classes sharing tier signature  $\mathbf{e}$ , stable under mild drift.
  - Spurious structure  $\Rightarrow \hat{R}(D)$  fails null-calibrated  $\tau_R$ , or classes are unstable / highly rewrite-heavy (low  $R_3$ ).
  - Cheating by rewrites  $\Rightarrow R_1, R_2$  high but  $R_3$  low; geometric mean suppresses acceptance.
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## Appendix A: Notes on topology operators and budgets

**Rewrite budgets** are explicit constraints, e.g.:

- max number of split/merge ops
- max JS divergence in degree/edge-size histograms
- max edit cost  $\Delta_{\text{edit}}$
- forbidden changes (e.g., preserve connected components)

These budgets can be enforced either as hard constraints (admissibility) or via  $R_3$  penalties.

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